

EMMA RAFKIN

(312) · 636 · 9908 ◇ emmarafkin@gmail.com ◇ emmarafkin.com

SUMMARY

NLP Research Scientist / Software Engineer with experience in applied AI and a focus in knowledge representation within LLMs. I am interested in making AI safer and expanding language technologies to speakers of all languages.

EDUCATION

Georgetown University

2023-2026

M.S. in Computational Linguistics

Overall GPA: 4.0

Master's Research Paper: Investigating the Cross-Lingual Transferability of NLP Tasks

Dartmouth College

2017-2021

B.A. in Computer Science

Minor in Linguistics

Overall GPA: 3.74, Cum Laude

EXPERIENCE

Johns Hopkins University Applied Physics Laboratory

August 2021 - Present

NLP Research Scientist/Software Engineer

Laurel, MD

- Created software for U.S. Government that inject domain knowledge into language models and extract critical information from text using state of the art NLP methods and Domain Ontologies.
- Awarded 8 grants to pursue research in AI transparency, concept erasure, summarization, entity disambiguation, LLM domain adaptation, AI red teaming, and multimodal anomaly detection
- Led research teams of 3-6 people, presented results to internal and external stakeholders

Dartmouth Applied Learning and Innovation (DALI) Laboratory

January 2018 - June 2021

Software Developer

Hanover, NH

- Produced full-stack web and mobile applications from scratch for external stakeholders every 10 weeks
- Created and led classes on web design and development
- Developer mentor, Women in Science Program lead, core executive board member for 2020-2021

Alarm.com

June 2020 - August 2020

Software Engineer Intern

Tysons, VA

- Created automated full-stack client-facing reports for property managers about all devices in their units.

Cambly Inc

January 2020 - March 2020

Software Engineer Intern

San Francisco, CA

- Upgraded Android application in Java and Kotlin to include English as a Second Language courses for adults

Dartmouth College Computer Science Department

September 2018 - December 2018

Teaching Assistant

Hanover, NH

- Taught weekly review class for Object Oriented Programming course to 10-15 students, graded coursework and tests

Thayer School of Engineering

January 2018 - June 2018

IT Help Desk Technician

Hanover, NH

- Answered computing questions for the students and faculty at Thayer

- Worked with clients to solve hardware difficulties. Implemented an organizational system across the computing services at the school.

Camp Nicolet for Girls

Camp Counselor

June 2017 - August 2018

Eagle River, WI

- Responsible for a cabin of 8 13-year-old girls for the 8-week season. Planned, supervised, and executed daily activities, meals, and special events
- Selected to lead a week-long advanced water ski skills camp. Led week long canoeing and hiking trips in the back country
- Certified in CPR, life guarding, and ski instruction.

FEATURED RESEARCH PROJECTS

Dandelion

Mechanistic Interpretability

JHU/APL

In progress

- Identifying mission-relevant negative LLM behavior during runtime, steering in activation & weight space to prevent that behavior from surfacing
- PI for a team of 3 researchers, won a year-long internal research grant

Using Task Arithmetic for Cross-Lingual Transfer in LLMs

Cross-lingual Task Transfer

Georgetown University

Under Review

- Explores the cross-lingual transfer of form & meaning tasks in LLMS as distance increases between source and target languages

Task Arithmetic with Support Languages for Low-Resource ASR

Low-Resource Language Modeling

Georgetown University

Awaiting Publication

- Incorporates information from genetically related high-resource languages to improve low-resource ASR models
- Preprint on ArXiv, awaiting publication in ACL Anthology

CaTE

Trustworthy AI development

JHU/APL

Complete

- Guidance for software engineers and project managers on data curation methods for trustworthy AI development
- Full report on ArXiv, separate paper on domain elicitation techniques published in 2024 International Conference on Assured Autonomy (ICAA).

TECHNICAL SKILLS

ML/NLP

Software Languages

Databases

CI/CD

Natural Languages

PyTorch, Hugging Face, fine-tuning & LoRA, activation probing & steering, RAG, Slurm

Python, JavaScript, TypeScript, React, Java, Kotlin

SQLite, PostgreSQL, ElasticSearch, MongoDB

Git, Docker

English (native), Spanish (conversational), ASL (beginner)